QQ-A-250/6G
September 21, 1978
SUPERSEDING
QQ-A-250/6F
December 4, 1970
(see 6.4)

FEDERAL SPECIFICATION

ALUMINUM ALLOY 5083, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 5083 plate and sheet; the general requirements are covered in QQ-A-250.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, H321, H323, H343 or H112 temper, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Philadelphia, Atlanta, Chicago, Kansas City, MO, Houston, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Magnesium	4.0	4.9
Manganese	0.40	1.0
Chromium	0.05	0.25
Silicon	-	0.40
Iron	-	0.40
Zinc	-	0.25
Titanium	-	0.15
Copper	-	0.10
Others, each	-	0.05
Others, total	-	0.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in this table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Mechanical properties

3.2.1 Mechanical properties of material as supplied. The mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

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TABLE II. Mechanical properties.

Temper	Thickness Inches	Tensile strength Minimum psi	Tensile strength Maximum psi	Yield strength 1/ Minimum psi	Yield strength 1/ Maximum psi	Elongation in 2 in. or 4 times D 2/, 3/, Minimum Percent
0	0.051 thru 1.500	40,000	51,000	18,000	29,000	16
	1.501 thru 3.000	39,000	50,000	17,000	29,000	16
	3.001 thru 4.000	38,000	-	16,000	-	16
	4.001 thru 5.000	38,000	-	16,000	-	14
	5.001 thru 7.000	37,000	-	15,000	-	14
	7.001 thru 8.000	36,000	-	14,000	-	12
H321	0.188 thru 1.500	44,000	56,000	31,000	43,000	12
	1.501 thru 3.000	41,000	56,000	29,000	43,000	12
H323	0.051 thru 0.125	45,000	54,000	34,000	44,000	8
	0.126 thru 0.249	45,000	54,000	34,000	44,000	10
H343	0.051 thru 0.125	50,000	59,000	39,000	49,000	6
	0.126 thru 0.249	50,000	59,000	39,000	49,000	8
H112	0.250 thru 1.500	40,000	-	18,000	-	12
	1.501 thru 3.000	39,000	-	17,000	-	12

1/ At 0.2 percent offset.
2/ Not required for material 1/2 inch or less in width
3/ D represents specimen diameter.

4. QUALITY ASSURANCE PROVISIONS (See QQ-A-250.)

5. PREPARATION FOR DELIVERY (See QQ-A-250.)

6. NOTES

6.1 Intended use. This alloy is for use where a weldable, moderate strength alloy having good corrosion resistance is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Form and temper required (see 1.2.1).

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(c) Dimensions required.
(d) Requirements for sizes not specifically covered (See QQ-A-250).
(e) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

6.3 Alloy 5083 should not be used for continuous service at temperatures exceeding 150°F because of susceptibility to stress corrosion cracking. In addition, stress corrosion susceptibility is increased by cold forming. Therefore, for critical applications bend radii shall not be less than ten times the thickness. For less critical applications bend radii shall not be less than five times the thickness. Forming at 425° ±25°F will avoid the adverse effect of cold work and high residual stresses, and it is recommended for this alloy.

6.4 Certain provisions of this specification are the subject of international standardization agreement ABC-NAVY-STD-44. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization officer, if required.

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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies of other documents referenced herein. Price 60 cents each.

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